THE UNIVERSITY OF ZAMBIA

SCHOOL OF NATURAL AND APPLIED SCIENCES

DEPARTMENT OF COMPUTING AND INFORMATICS

Final Year Project Proposal

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Project Title: MediVault: A Role Based Patient Portal for the University of Zambia Clinic

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1. Introduction

Healthcare information management is a very important part of ensuring that patients receive the right care. In universities where there is a large number of students, the likelihood of having a student with a medical condition or health related incident is higher. Because of this, the ability of a university to quickly verify the identity of a patient and access accurate medical histories can improve the quality of the treatment they receive. Despite the development of several modern solutions in healthcare, many university clinics, especially in developing countries have continued to rely on physical identification to verify patients and physical information storage.

This is how The University of Zambia Clinic currently operates. Students are required to present their physical university identity cards to verify that they are eligible to receive medical care from the clinic. Any existing medical histories of the patient are not readily accessible to any medical personnel that is attending to the patient. There is no system in place for situations in which an unconscious patient is brought into the clinic and is not in a state to give their consent to receive treatment and there is no way to ensure medical personnel at the clinic are aware of any pre-existing conditions the patient may have, medication they might be taking, or any known allergies before they are given any kind of treatment.

The problems we have mentioned describe real risks to patient safety and how efficient the clinic operates. The lack of a digital system means that the quality of care that a student receives will be dependent on what information is available in that moment, information which may be incomplete, outdated or missing altogether.

Our system, MediVault will be designed to address these changes by providing a secure, web based patient portal which will be specifically for the University of Zambia Clinic. The system will digitise and centralise all student medical records, replace physical identity verification with student number lookup and introduce a structured model that will ensure that only authorised personnel can view or modify sensitive patient information. An additional feature that allows patients to assign a next of a kin who can give consent for them to receive care in a situation where they are physically unable to give consent themselves, a gap which is currently not addressed by the record keeping the clinic has in place.

1.1. Problem Statement

The University of Zambia Clinic currently operates without a centralized and secure digital system for managing student patient records, which creates several challenges:

- Students are required to physically present their university ID cards to confirm their enrolment before receiving care. This reliance on physical documents can be problematic when cards are lost, forgotten, or simply unavailable.
- In addition, doctors do not always have access to a patient’s medical history at the point of care. As a result, treatment decisions may be made without full knowledge of existing conditions, medications, or allergies.
- Another major concern is the lack of a formal process for accessing patient information or obtaining consent when a patient is physically unable to give consent. This becomes especially critical in emergency situations where timely decisions are required.
- There is also no structured method to monitor who accesses or modifies patient records, which raises concerns about accountability and the integrity of the data.

Overall, these issues reduce the quality and safety of care, increase administrative workload, and leave patient data vulnerable to uncontrolled access.

1.2. Project Aim

The goal is to design and build MediVault, a secure, role based web portal for the UNZA clinic. The system will digitise student medical records, remove the need for physical ID checks by allowing verification through student numbers, and introduce clear access controls along with delegated consent features. Together, these improvements aim to make patient care safer, more efficient, and more reliable.

1.3. Objectives

1.3.1. Design and implement a role based access control system, with clear permissions assigned to patients, receptionists, doctors, and administrators.
1.3.2. Develop a student verification feature that allows receptionists to confirm eligibility using a student number, removing the need for physical ID cards.
1.3.3. Create a structured electronic medical profile for each patient, including diagnoses, medications, allergies, and visit history, so authorised clinical staff can easily access it during care.
1.3.4. Add a next of kin consent feature that lets patients pre-authorise a trusted person to make decisions on their behalf if they become incapacitated.
1.3.5. The system will include a simple logging mechanism that records when patient records are accessed or updated, along with the user responsible. This will provide a basic level of accountability and transparency without introducing unnecessary complexity

1.4. Scope of the Project

In Scope:

- A web based patient portal that can be accessed through standard browsers on both desktop and mobile devices.
- Role based access for five types of users: students (patients), receptionists, doctors, administrators, and next of kin.
- Student number verification to replace the need for physical ID cards.
- Structured electronic medical profiles covering diagnoses, medications, allergies, and visit history.
- A next of kin setup with a delegated consent process.
- Patient records will be stored securely using standard web and database security practices, such as controlled access, authentication, and HTTPS for data transmission.
- The system will also include a basic access log that records key actions, such as when records are viewed or modified and by which user.
- An admin dashboard for managing user accounts and monitoring access.

Out of Scope:

- Integration with external hospital systems or national health platforms.
- Native mobile apps (iOS or Android), as the system will be web only.
- Billing, payment processing, or insurance related features.
- Telemedicine or remote consultation functionality.
- Any deployment beyond the UNZA clinic.

- The project will not include containerisation technologies such as Docker, as the system will be run locally for demonstration purposes.
- Advanced encryption mechanisms, such as custom implementations of AES-256, are also outside the scope of this project and may be considered as future improvements.
- Additionally, features that allow patients to manually grant or revoke access to their records will not be implemented, as access control will be fully handled through predefined user roles.

2. Literature Overview

Patient portal systems and electronic health record (EHR) platforms have been widely researched and are increasingly being used in both developed and developing healthcare systems. This section reviews key studies and technologies that are relevant to the development of MediVault.

Electronic Health Records and Patient Portals

The transition from paper based records to digital systems has been well documented, with many studies highlighting improvements in healthcare delivery. Digital records help reduce medical errors, improve the quality of care, and make it easier for healthcare providers to access patient information when needed.

Patient portals, which allow individuals to view their own health records online, have become an important part of modern EHR systems. Research by Ammenwerth et al. shows that when patients can access their own medical information, they are more likely to follow treatment plans and communicate more effectively with healthcare providers. MediVault builds on this idea by including features that allow patients to directly access their records.

Role Based Access Control in Healthcare Systems

Role Based Access Control (RBAC) is commonly used in healthcare systems to manage who can access specific types of information. It is based on the principle that users should only have access to the data necessary for their role.

In practice, this means that administrative staff may handle scheduling and general information but cannot view sensitive clinical data, while healthcare professionals can access and update medical records but not administrative data. This approach is reflected in MediVault’s design, which uses RBAC to ensure that data access is controlled and secure. The model proposed by Ferraiolo et al. provides the foundation for this structure.

Delegated Consent and Proxy Access

Another important concept in patient portal systems is delegated or proxy access. This allows a trusted individual to access a patient’s information or make decisions on their behalf when the patient is unable to do so.

This approach is commonly used in cases involving children or elderly patients. MediVault adapts this idea to a university clinic environment, where students may temporarily be unable to give consent due to illness or injury. In such cases, a designated proxy can assist in managing their healthcare needs.

Digital Health in Sub Saharan African University Settings

Although research in this area is still developing, there is growing interest in the use of digital health systems within sub Saharan Africa, particularly in academic institutions. Similarly, Chibawe and Nyirenda (2026) demonstrate the potential of integrated digital systems to improve health outcomes.

These studies show that implementing systems like MediVault in a university setting is both relevant and possible, especially as digital infrastructure continues to improve in the region.

Data Encryption and Security in Healthcare Applications

Because medical data is highly sensitive, strong security measures are essential in any healthcare system. Many international standards, such as those aligned with HIPAA and GDPR, recommend the use of AES-256 encryption for stored data and TLS for data transmitted over networks.

Although Zambia does not yet have a data protection framework as comprehensive as GDPR, it is still important to follow globally accepted best practices. In this project, security will be approached using standard web technologies such as HTTPS and controlled database access, while more advanced encryption techniques will be considered as potential future enhancements.

3. Proposed Methodology

Development Approach

The project will follow an Agile inspired approach, where the system is built in small, manageable parts rather than all at once. A modular, feature by feature approach will be used, meaning each feature (e.g., viewing medical records) will include its interface, backend logic, and database interaction.

Version control will be handled using GitHub, allowing team members to work on different parts of the system simultaneously.

System Architecture / Workflow

The system will use a full stack architecture based on Next.js.

Users interact with the frontend, which sends requests to backend API routes. The backend processes the request, applies role based access control, and retrieves or updates data in the PostgreSQL database. The response is then returned and displayed to the user.

The system will be organised into modules such as authentication, medical records, and next of kin consent.

Data Collection

No real patient data will be used. Instead, fake data will be created to simulate students, medical records, and user roles for testing and demonstration purposes.

3.1.Tools and Technologies

Category	Technology
Programming Languages	HTML, CSS, JavaScript
Frameworks/Libraries	Next.js (for both frontend and backend), JSON Web Tokens (JWT) for authentication
Databases	PostgreSQL
Development Environment	Visual Studio Code, GitHub, pgAdmin

3.2. Ethical and Legal Considerations

Data Privacy

Medical records are highly sensitive, so protecting patient information is a key concern in this project. Data will be secured using standard practices such as authenticated access, role based restrictions, and HTTPS to protect information during transmission. These measures help ensure that only authorised users can access patient data while keeping the system practical to implement.

User Consent

Before using the system, patients will be required to give informed consent during registration. This process will clearly explain how their data will be stored and who may have access to it. For features such as next of kin or proxy access, the patient must explicitly assign a trusted individual. This ensures that no one is given access to a patient’s information without their direct approval.

Security

To protect the system from potential threats, a number of standard security measures will be implemented. These include secure password hashing, JWT based session handling, and the use of HTTPS to protect data in transit. Input validation will also be applied to prevent common attacks such as SQL injection, cross site scripting (XSS), and cross site request forgery (CSRF). Together, these measures help ensure that the system remains secure and reliable.

Intellectual Property

All code developed for MediVault will be original and created by the project team. Open source tools and frameworks may be used where appropriate, in line with their licences. No real patient data will be used during development or testing; instead, synthetic data will be generated to simulate realistic scenarios while maintaining ethical standards.

4. Expected Outcomes

- By the end of the project, MediVault is expected to deliver a complete and functional web based patient portal that can be used within the UNZA clinic environment. The system will be fully deployed and accessible to its intended users.
- One of the key outcomes is the implementation of role based access control, supporting five different user types: patients, receptionists, doctors, administrators, and next of kin. Each role will have clearly defined permissions to ensure that users can only access information relevant to their responsibilities.
- The system will also include a student identity verification feature, which will reduce or eliminate the need for physical ID cards during clinic visits. In addition, each patient will have a structured electronic medical profile, making it easier for healthcare providers to access and manage health information efficiently.
- Another important feature is the next of kin functionality, which will allow patients to assign a trusted individual who can act on their behalf when necessary. This will be supported by a delegated consent workflow to ensure that access is always properly authorised.
- To protect sensitive information, patient records will be stored securely using established web and database security practices, ensuring that access is properly controlled and protected. The system will also include a simple logging feature that records when records are accessed or updated, helping to provide basic accountability within the system.
- In terms of documentation, the project will include detailed technical materials such as system architecture diagrams, API documentation, and user guides tailored to each type of user. Finally, a comprehensive project report will be produced, outlining the design choices, implementation process, testing outcomes, and key reflections from the development experience.

5. Project Timeline

Phase	**Duration**
Proposal	Week 1-4
Literature Review & System Design	Week 5-7
Initial Development	Week 8-9
Core System Development	Week 10-13
Midway Evaluation	Week 14
Testing and Final Development	Week 15-19
Final Report Writing	Week 20-23
Final Submission	Week 24

6. Risk Analysis

Risk	**Mitigation Strategy**
Access to real student data for testing	Use synthetically generated test data throughout development; request anonymised sample data only for final validation if approved.
Scope creep beyond clinic context	Adhere strictly to the defined scope; document any supervisor approved scope changes formally.

Security vulnerabilities in the portal	Follow basic security best practices and perform simple security testing during development.
Team member unavailability	Maintain shared code repository on GitHub; document all progress in regular sprint notes.
Technology compatibility issues	Use widely supported, LTS versions of all frameworks; validate choices early in the setup phase.
Insufficient user testing participants	Engage fellow students early to recruit testers; conduct testing iteratively across multiple sprints.
Authentication or session handling issues (JWT)	Test authentication flows thoroughly and validate token handling during development.

7. References

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Supervisor Approval

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